

IF332

FORMAL LANGUAGES

&

AUTOMATA THEORY

05
MINIMIZATION OF DFA

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REVIEW

Epsilon-NFA (ϵ -NFA):

Epsilon-NFA (ϵ -NFA)

Epsilon-Closures

Conversion of ϵ -NFA to NFA

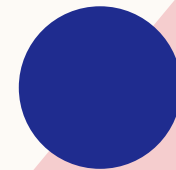
Union and Concatenation of FSA

OUTLINE

Minimization of DFA

Partitioning Method

Table Filling Method (Myhill Nerode Theorem)



MINIMIZATION OF DFA

- Finding an equivalent DFA that has as few states as any DFA accepting the same language

EQUIVALENCE OF STATES

- Two states p and q are said to be EQUIVALENT or INDISTINGUISHABLE if:
 - $\delta(p, w) \in F$ and $\delta(q, w) \in F$
 - $\delta(p, w) \notin F$ and $\delta(q, w) \notin F$
- If $|w| = 0$, then p and q are said to be 0 equivalent
If $|w| = 1$, then p and q are said to be 1 equivalent
If $|w| = 2$, then p and q are said to be 2 equivalent
If $|w| = n$, then p and q are said to be n equivalent
- If we find that states p and q are equivalent, and we also find that q and r are equivalent, then it must be that p and r are equivalent
 - Therefore, the three states are equivalent

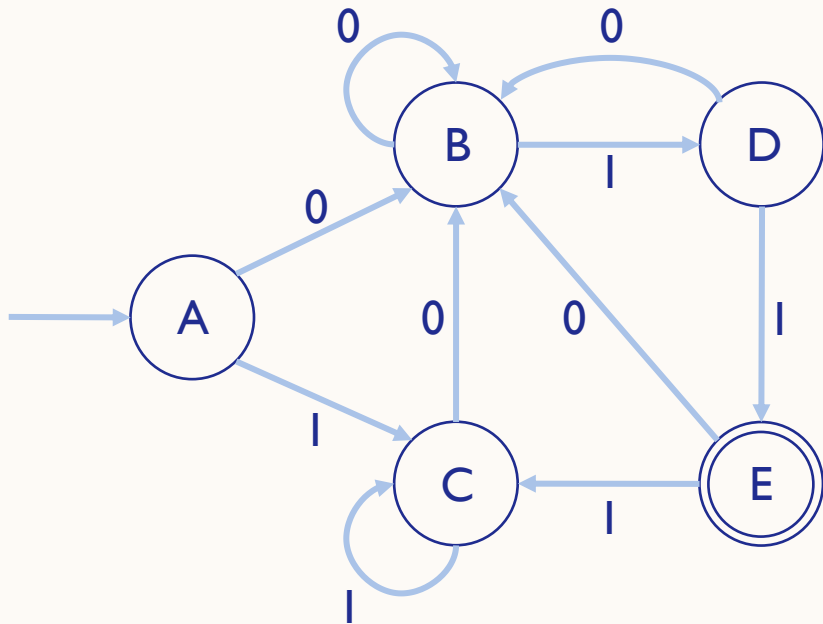
EQUIVALENCE OF STATES

- Two states p and q are said to be DISTINGUISHABLE if:
 $\delta(p, w) \notin F$ and $\delta(q, w) \in F$

MINIMIZATION OF DFA

PARTITIONING METHOD

Minimize the following DFA.



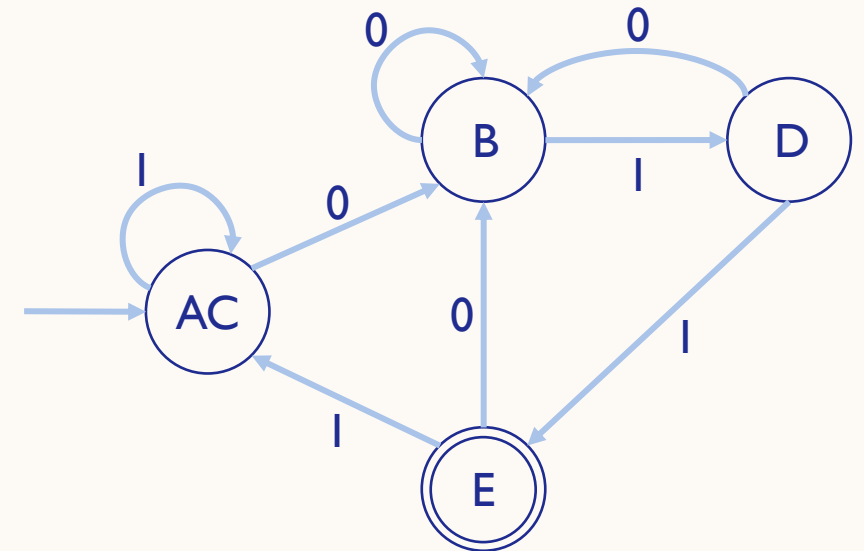
δ	0	1
$\rightarrow A$	B	C
B	B	D
C	B	C
D	B	E
$*E$	B	C

0 Equivalence: $\{A, B, C, D\}$ $\{E\}$

1 Equivalence: $\{A, B, C\}$ $\{D\}$ $\{E\}$

2 Equivalence: $\{A, C\}$ $\{B\}$ $\{D\}$ $\{E\}$

3 Equivalence: $\{A, C\}$ $\{B\}$ $\{D\}$ $\{E\}$

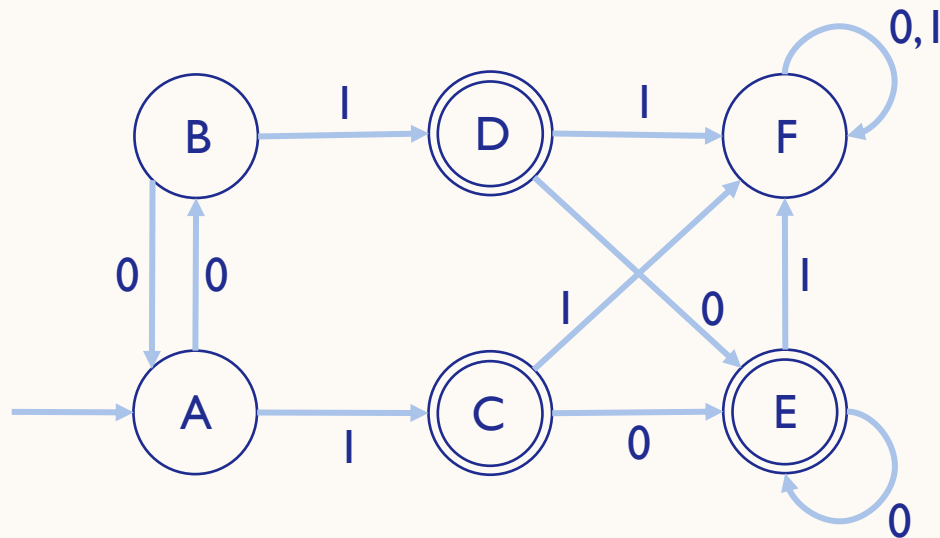


MINIMIZATION OF DFA

PARTITIONING METHOD

WITH MULTIPLE FINAL STATES

Minimize the following DFA.



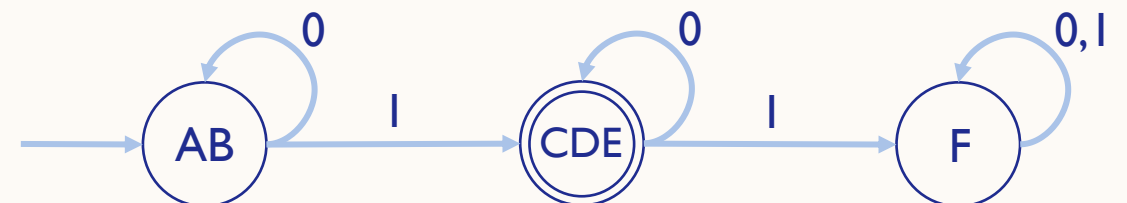
δ	0	1
$\rightarrow A$	B	C
B	A	D
*C	E	F
*D	E	F
*E	E	F
F	F	F

0 Equivalence: {A, B, F} {C, D, E}

1 Equivalence: {A, B} {F} {C, D, E}

2 Equivalence: {A, B} {F} {C, D, E}

δ	0	1
$\rightarrow \{A, B\}$	{A, B}	{C, D, E}
{F}	{F}	{F}
*{C, D, E}	{C, D, E}	{F}

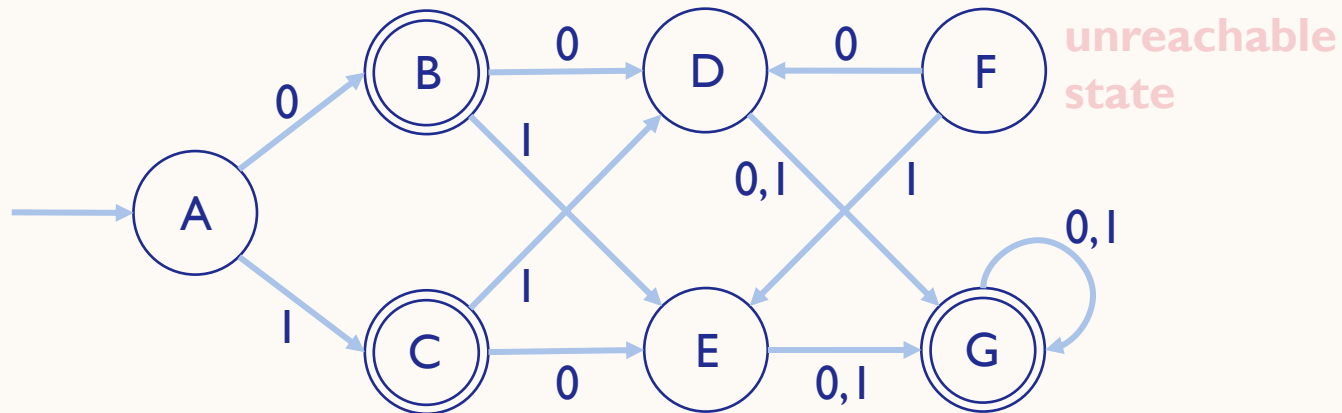


MINIMIZATION OF DFA

PARTITIONING METHOD

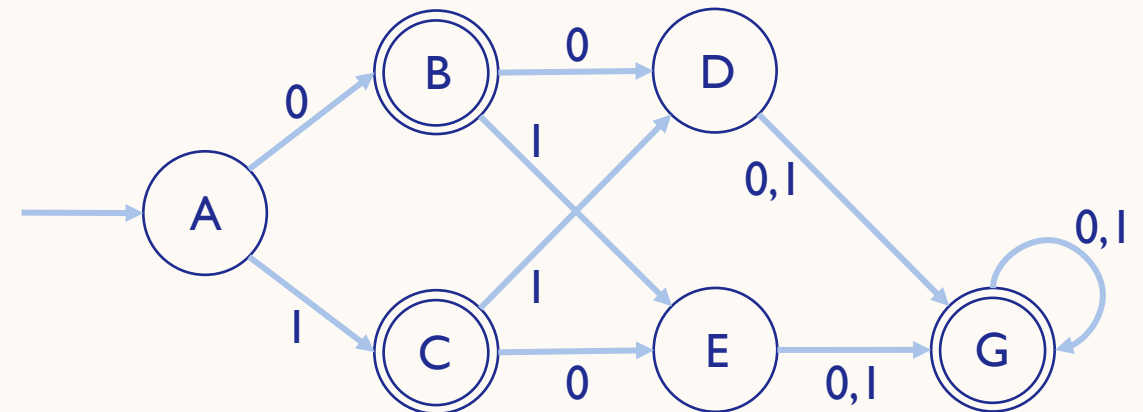
WITH UNREACHABLE STATES

Minimize the following DFA.



A state is said to be unreachable if there is no way it can be reached from the initial state

Eliminate any state that cannot be reached from the start state

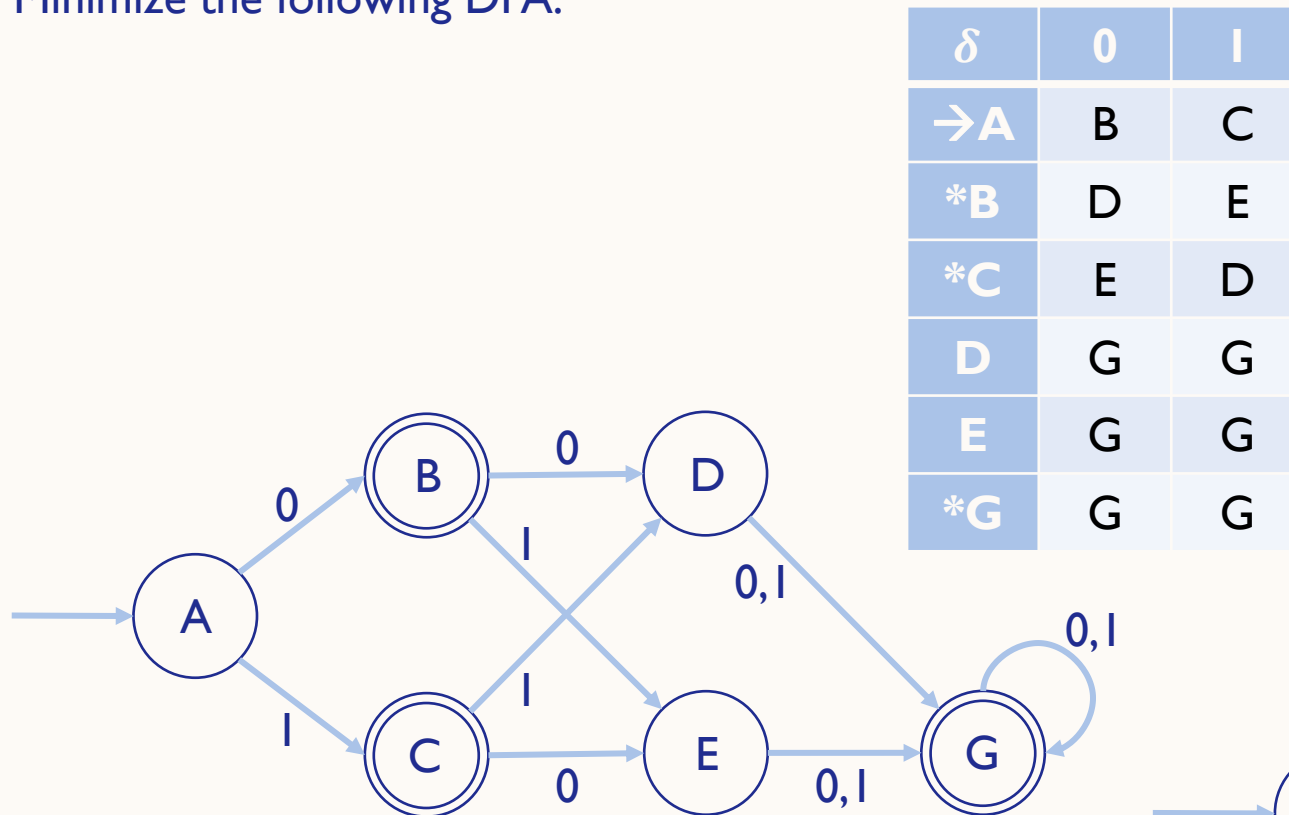


MINIMIZATION OF DFA

PARTITIONING METHOD

WITH UNREACHABLE STATES

Minimize the following DFA.



0 Equivalence: $\{A, D, E\}$ $\{B, C, G\}$

1 Equivalence: $\{A, D, E\}$ $\{B, C\}$ $\{G\}$

2 Equivalence: $\{A\}$ $\{D, E\}$ $\{B, C\}$ $\{G\}$

3 Equivalence: $\{A\}$ $\{D, E\}$ $\{B, C\}$ $\{G\}$

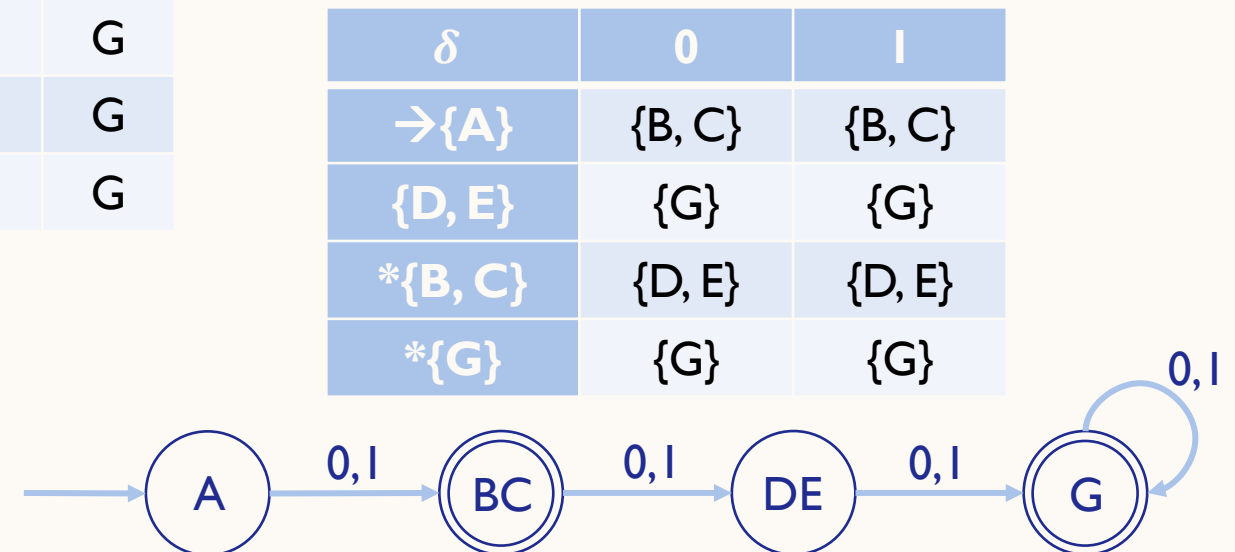


TABLE FILLING METHOD (MYHILL NERODE THEOREM)

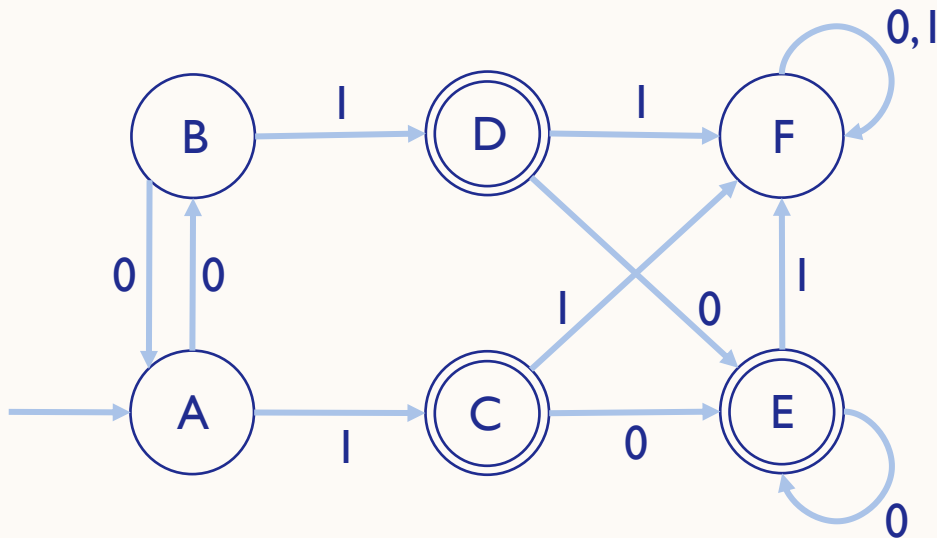
1. Draw a table for all pairs of states (p, q)
2. Mark all pairs where $p \in F$ and $q \notin F$
3. If there are any unmarked pairs (p, q) such that $[\delta(p, w), \delta(q, w)]$ is marked, then mark $[p, q]$; where w is an input symbol
 - REPEAT THIS UNTIL NO MORE MARKINGS CAN BE MADE
4. Combine <all> the unmarked pairs and make them a <single> state in the minimized DFA

MINIMIZATION OF DFA

TABLE FILLING METHOD

(MYHILL NERODE THEOREM)

Minimize the following DFA.

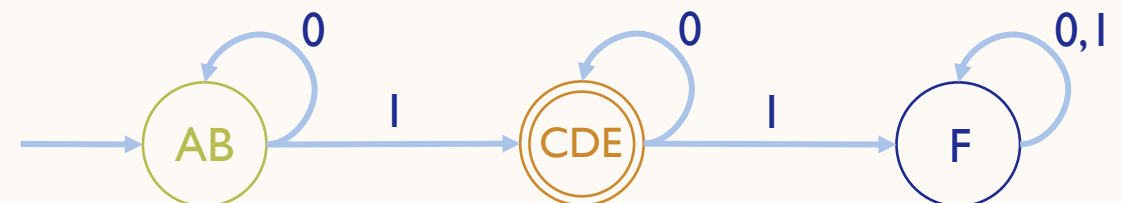


	A	B	C	D	E	F
A						
B						
C	✓	✓				
D	✓	✓				
E	✓	✓				
F	✓	✓	✓	✓	✓	

States	0	1
(B,A)	(A,B)	(D,C)
(D,C)	(E,E)	(F,F)
(E,C)	(E,E)	(F,F)
(E,D)	(E,E)	(F,F)
(F,A)	(F,B)	(F,C)
(F,B)	(F,A)	

Unmarked:

(B,A) (D,C) (E,C) (E,D)

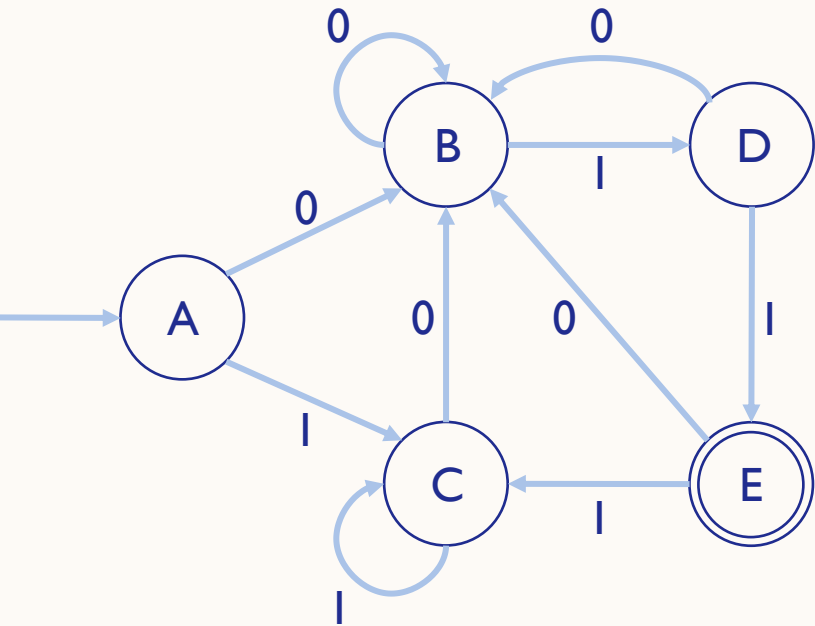


MINIMIZATION OF DFA

TABLE FILLING METHOD

(MYHILL NERODE THEOREM)

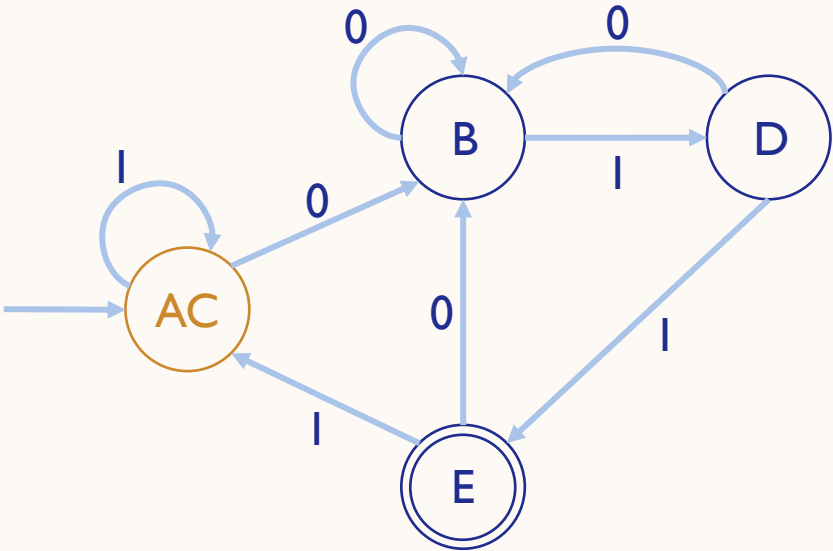
Minimize the following DFA.



Unmarked:
(C,A)

States	0	1
(B,A)	(B, B)	(D, C)
(C,A)	(B, B)	(C, C)
(C,B)	(B, B)	(C, D)
(D,A)	(B, B)	(E, C)
(D,B)	(B, B)	(E, D)
(D,C)	(B, B)	(E, C)
(B,A)	(B, B)	(D, C)
(C,A)	(B, B)	(C, C)
(C,B)	(B, B)	(C, D)
(C,A)	(B, B)	(C, C)

	A	B	C	D	E
A					
B	✓				
C		✓			
D	✓	✓	✓		
E	✓	✓	✓	✓	

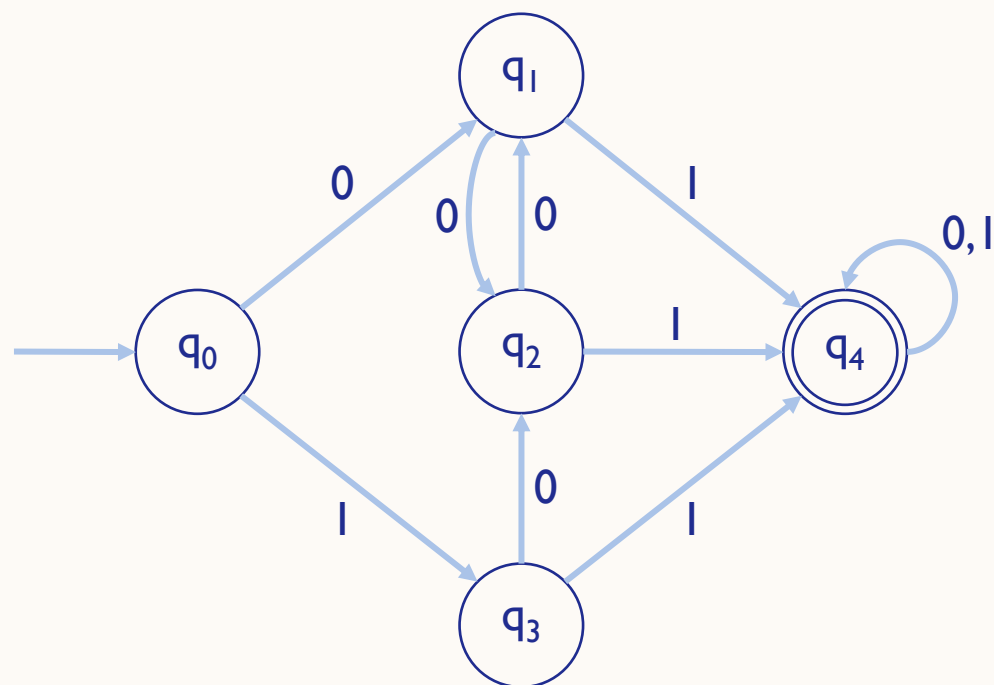




PRACTICE

EXERCISES

1. Minimize the following DFA.



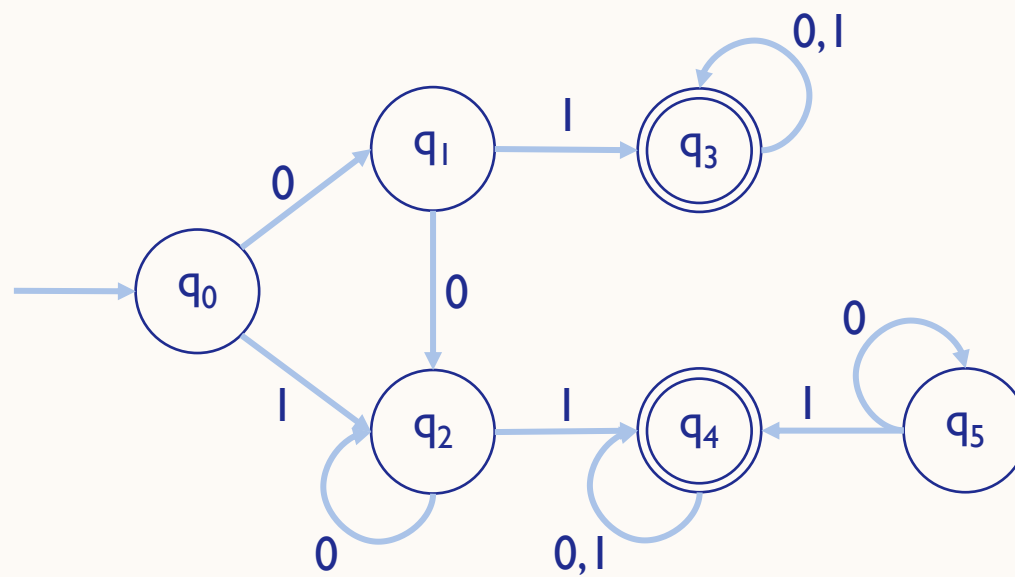
EXERCISES

2. Construct a minimum DFA equivalent to the DFA described below.

δ	0	1
$\rightarrow q_0$	q_1	q_5
q_1	q_6	q_2
$*q_2$	q_0	q_2
q_3	q_2	q_6
q_4	q_7	q_5
q_5	q_2	q_6
q_6	q_6	q_4
q_7	q_6	q_2

EXERCISES

3. Minimize the following DFA.



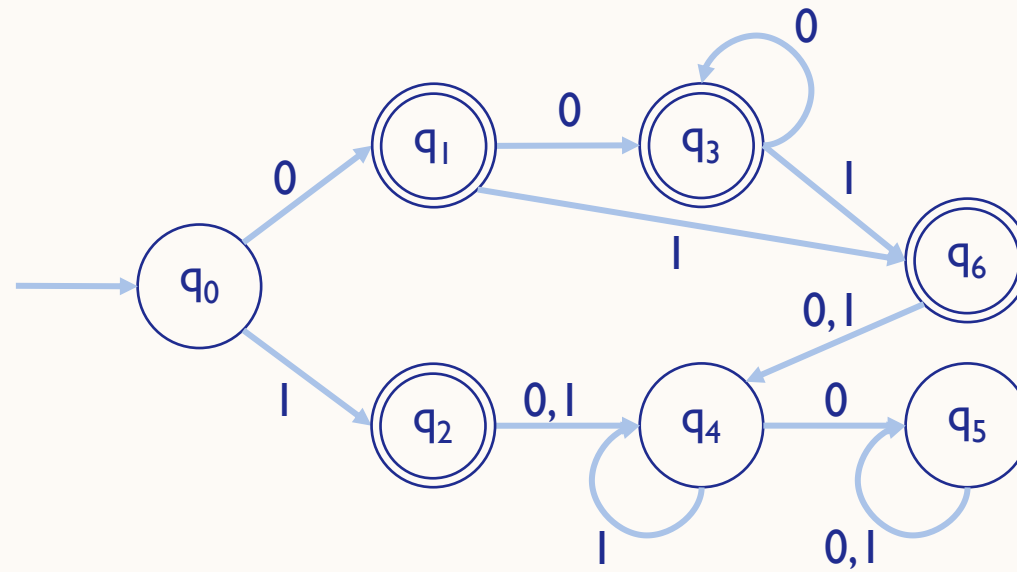
EXERCISES

4. Construct the minimum-state equivalent DFA.

δ	0	1
$\rightarrow A$	B	E
B	C	F
*C	D	H
D	E	H
E	F	I
*F	G	B
G	H	B
H	I	C
*I	A	E

EXERCISES

5. Minimize the following DFA.



REFERENCES

- Utdirartatmo, F. (2005), Teori Bahasa dan Otomata (edisi ke-2), Graha Ilmu.
- Hopcroft, John E., Rajeev Motwani, and Jeffrey D. Ullman (2006), Introduction to Automata Theory, Languages, and Computation (3rd edition), Addison-Wesley.

NEXT

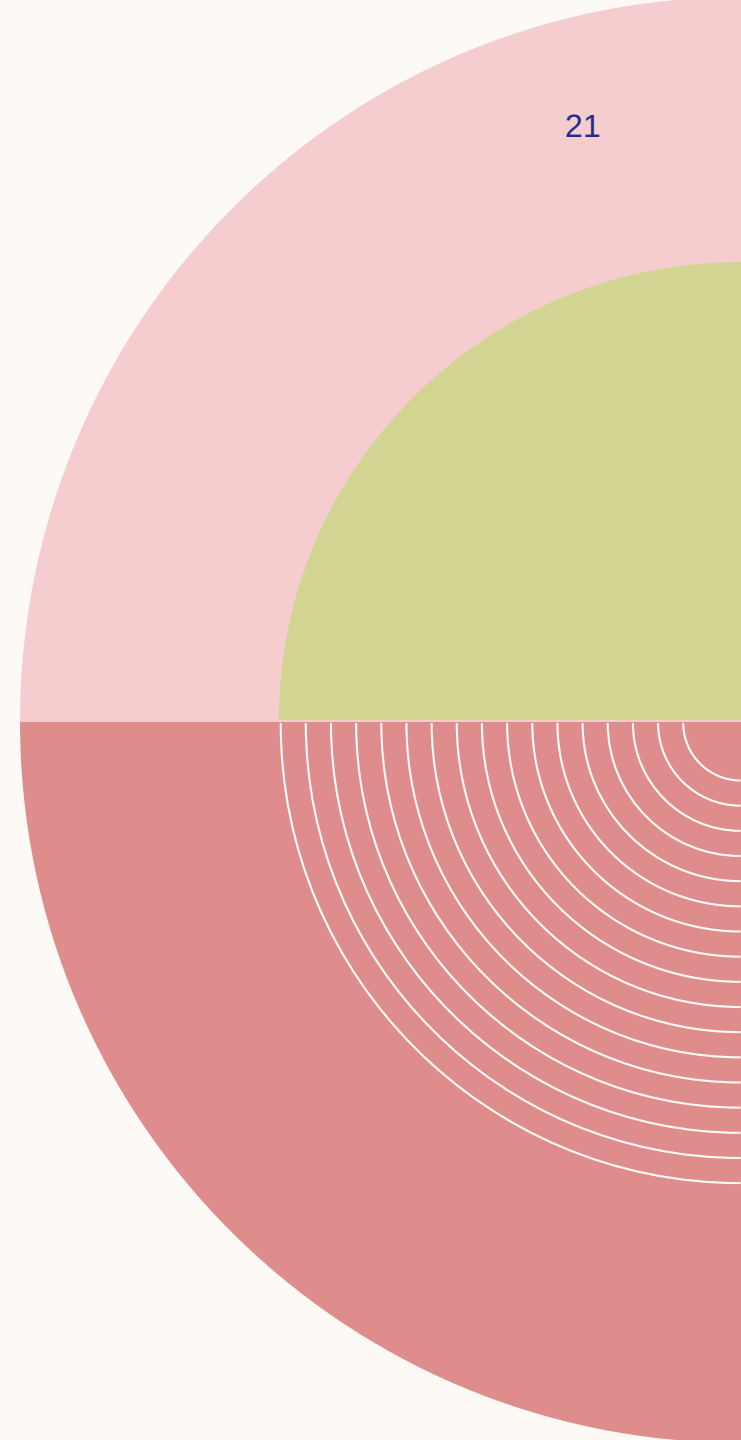
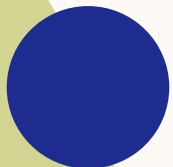
Finite Automata with Outputs:

Mealy Machine

Moore Machine

Conversion of Moore Machine to Mealy Machine

Conversion of Mealy Machine to Moore Machine



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To become an **outstanding** undergraduate Computer Science program that produces **international-minded** graduates who are **competent** in software engineering and have **entrepreneurial spirit** and **noble character**.

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3. To deliver science-and-technology-based society services to implement science and technology



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